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Part 30 – User defined functions

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In this session we will learn

- User Defined Functions
- Types of User Defined Functions
- Creating a Scalar User Defined Function
- Calling a Scalar User Defined Function
- Places where we can use Scalar User Defined Function
- Altering and Dropping User Defined Function

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SCALAR UDF

From **Parts 22 to 29**, we have learnt how to use many of the **system functions** that are available in SQL Server. In this session, we will turn our attention, to creating **user defined functions**. In short UDF.

In SQL Server there are 3 types of User Defined functions

1. Scalar functions
2. Inline table-valued functions
3. Multi-statement table-valued functions

Scalar functions may or may not have parameters, but always return a single (scalar) value. The returned value can be of any data type, except **text, ntext, image, cursor, and timestamp**

--To create a function, we use the following syntax:

```
CREATE FUNCTION Function_Name(@Parameter1 DataType, @Parameter2 DataType, ..., @ParameterN Datatype)
RETURNS Return_Datatype
AS
BEGIN
    --Function Body
    Return Return_Datatype
END
```

SCALAR UDF

```
CREATE FUNCTION Age (@DOB Date)
RETURNS INT
AS
BEGIN
    DECLARE @Age INT
    SET @Age = DATEDIFF(YEAR, @DOB, GETDATE()) -
        CASE
            WHEN (MONTH(@DOB) > MONTH(GETDATE())) OR
                 (MONTH(@DOB) = MONTH(GETDATE()) AND DAY(@DOB) > DAY(GETDATE()))
            THEN 1
            ELSE 0
        END
    RETURN @Age
END
```

When calling a scalar user-defined function, you must supply a two-part name, **OwnerName.FunctionName**. **dbo** stands for database owner.

```
Select dbo.Age('10/08/1982')
```

You can also invoke it using the complete 3 part name, **DatabaseName.OwnerName.FunctionName**

```
Select SampleDB.dbo.Age(dbo.Age('10/08/1982'))
```

SCALAR UDF

```
--Scalar user defined functions can be used in the Select clause  
Select Name, DateOfBirth, dbo.Age(DateOfBirth) as Age from tblEmployees
```

Id	Name	DateOfBirth
1	Sam	1980-12-30 00:00:00.000
2	Pam	1982-09-01 12:02:36.260
3	John	1985-08-22 12:03:30.370
4	Sara	1979-11-29 12:59:30.670

Name	DateOfBirth	Age
Sam	1980-12-30 00:00:00.000	31
Pam	1982-09-01 12:02:36.260	30
John	1985-08-22 12:03:30.370	27
Sara	1979-11-29 12:59:30.670	32

```
--Scalar user defined functions can be used in the Select Where clauses  
Select Name, DateOfBirth, dbo.Age(DateOfBirth) as Age from tblEmployees  
Where dbo.Age(DateOfBirth) > 30
```

Id	Name	DateOfBirth
1	Sam	1980-12-30 00:00:00.000
2	Pam	1982-09-01 12:02:36.260
3	John	1985-08-22 12:03:30.370
4	Sara	1979-11-29 12:59:30.670

Name	DateOfBirth	Age
Sam	1980-12-30 00:00:00.000	31
Sara	1979-11-29 12:59:30.670	32

A stored procedure also can accept **DateOfBirth** and **return Age**, but you cannot use stored procedures in a select or where clause. This is just one difference between a function and a stored procedure. There are several other differences, which we will talk about in a later session.

To alter a function we use **ALTER FUNCTION FuncationName** statement and to delete it, we use **DROP FUNCTION FuncationName**.

Additional Resources

- PRAGIM Home Page:

- <http://www.PragimTech.com>

- Resources:

- ASP.NET Interview Questions
 - <http://www.VenkatASPInterview.Blogspot.com>
 - C# Interview Questions
 - <http://www.VenkatCSharpInterview.Blogspot.com>

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